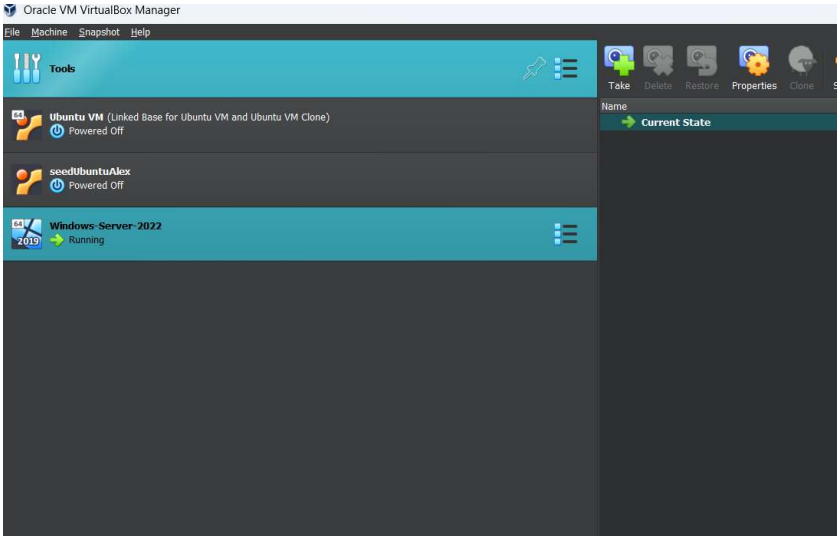
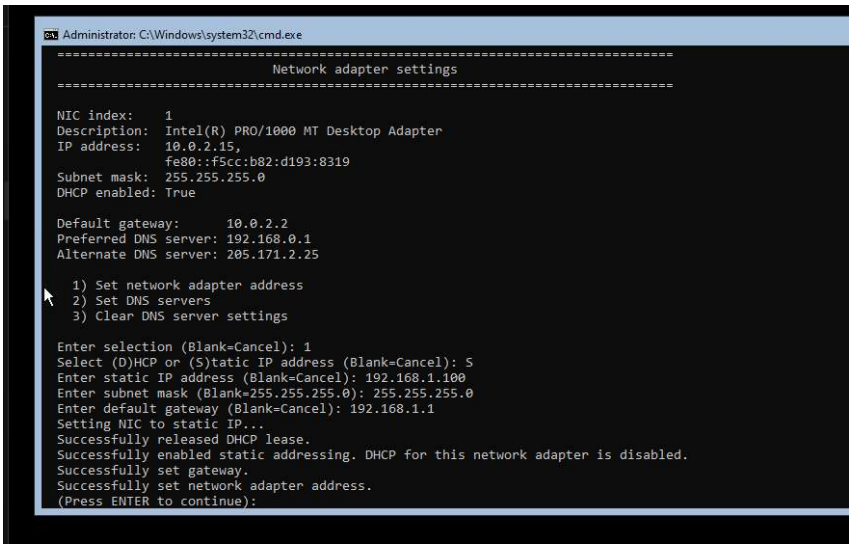
Day 1 Setup Windows Environment and Windows Server-2022

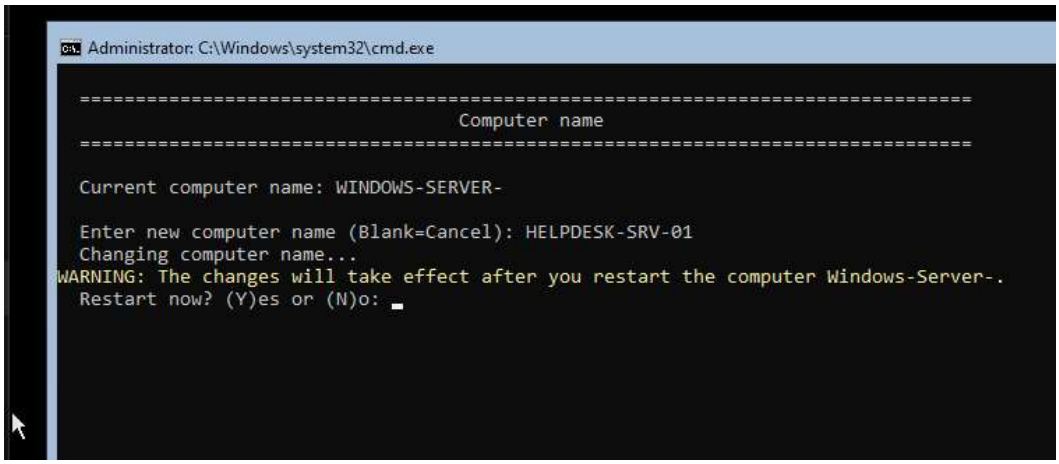
- Restarted my PC, entered BIOS/UEFI, and confirmed Virtualization was enabled
- Installed/Configured a VirtualBox VM on my windows PC
- Downloaded a server ISO and client ISO both windows (Windows Server 2022 and Windows 11 ISO), configured them, and ran their executables
- Created a Windows VM, mounted just downloaded Windows ISO, and installed the windows server inside of the VM



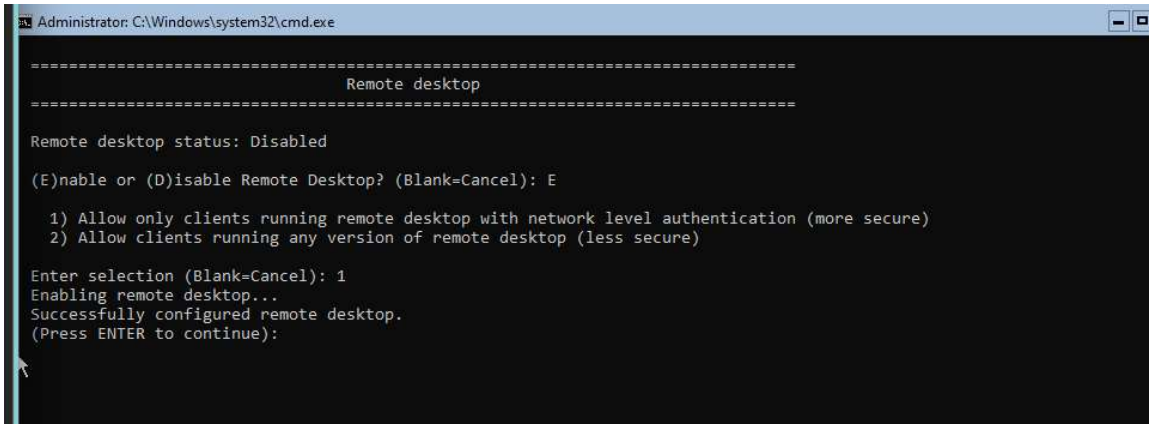
- Updated network adapter settings specifically the static IP address, the subnet mask, and the default gateway. This might be changed later.



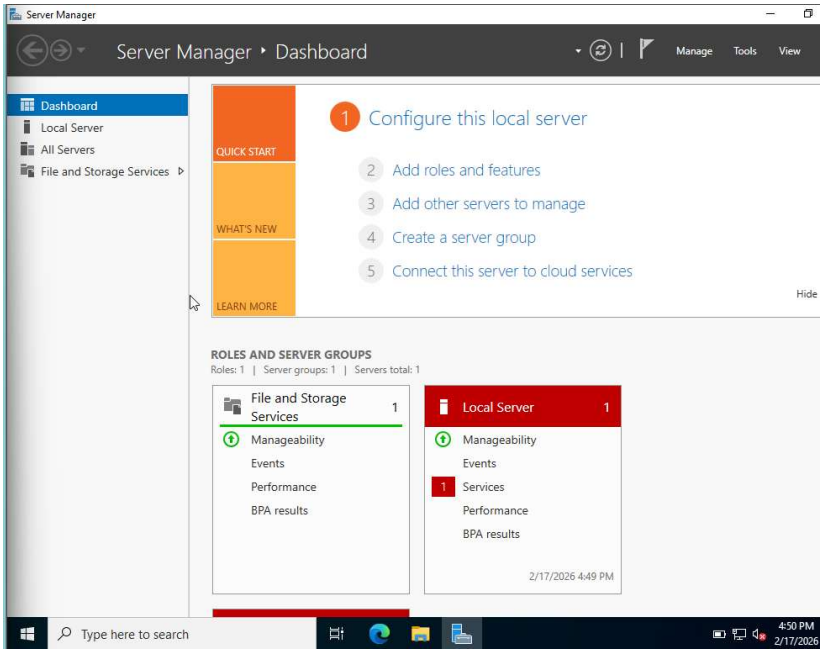
- Renamed computer for its specific purpose (help desk)



- Updated for allowing the use of RDP to this server but only if they have network level authentication for security purposes

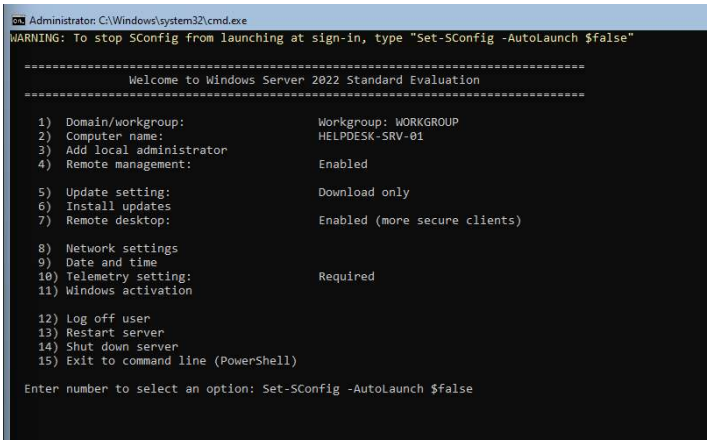


- Ensured that the server manager could be properly accessed so its ready for day 2



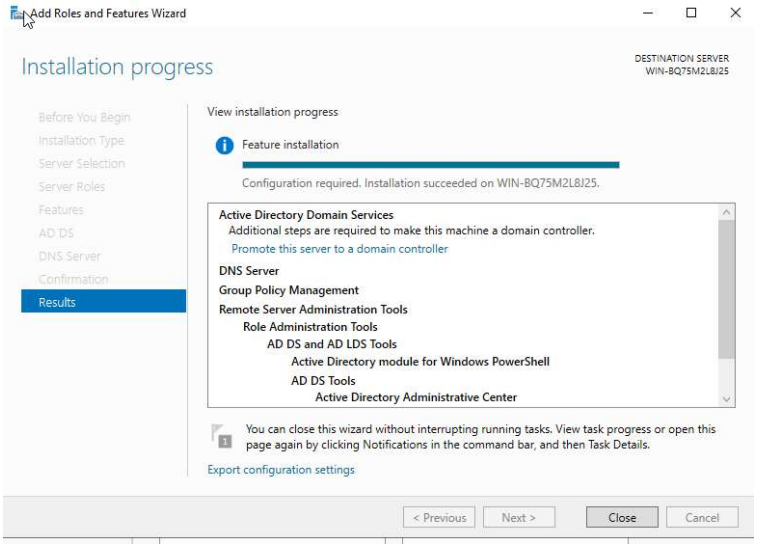
Problems/Observations from Day 1

- The big issue I had was I added the wrong ISO file when I initially setup the server (Windows 10 instead of Windows Server 2022), so I had to remove the first VM, and reconfigure it
- The good this is that I noticed this issue right away when the server manager was now popping up
- When I tried to restart the server after adjusting the computer name, remote desktop, and network settings, it kept bringing me back to the administrator command prompt and not starting the server properly. So, I had to trouble shoot that issue by moving to the command prompt and then forcing it to restart.

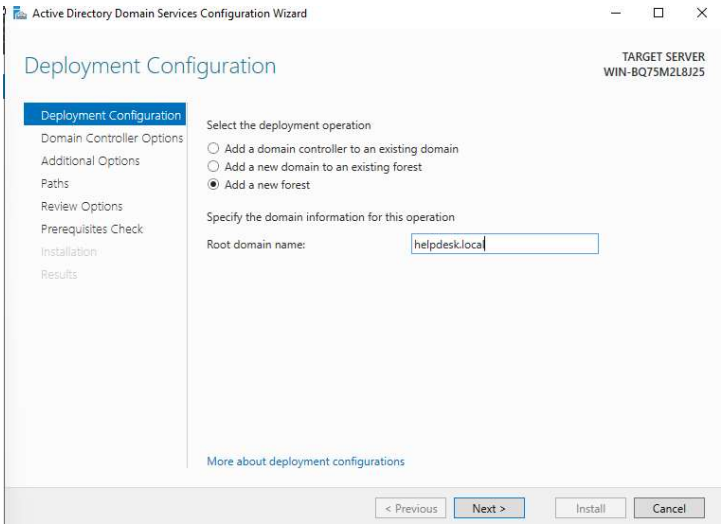


Day 2 Setting up Active Directories

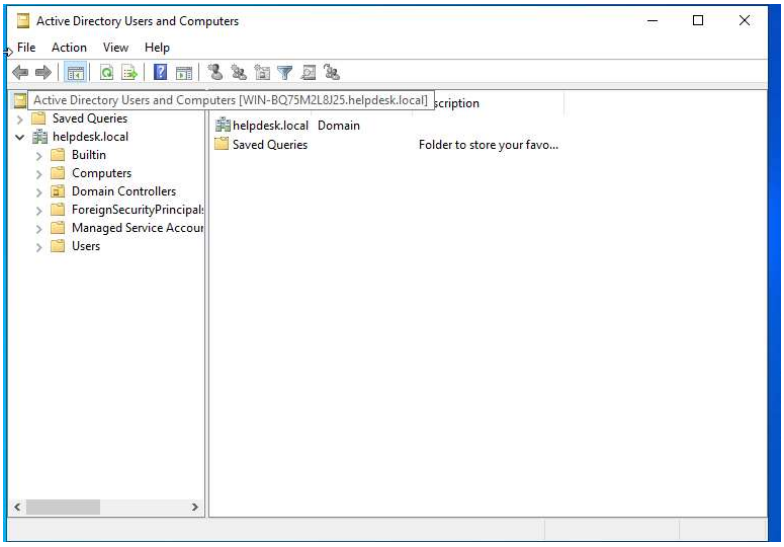
- Installed Active Directory Domain Services as well as DNS Server



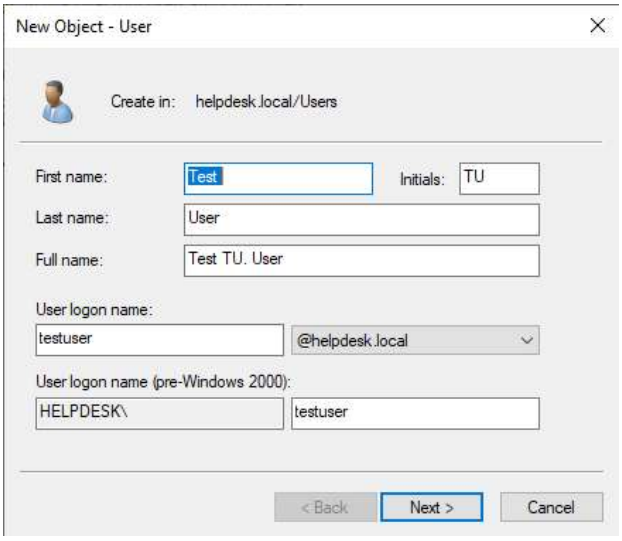
- Promoted server to be able to be a domain controller

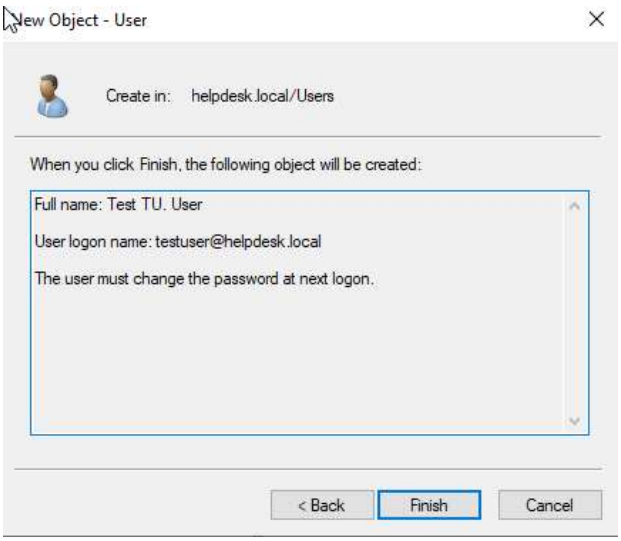


- Double checked after VM restarted to ensure that AD was properly setup, and has the proper built in containers



- Created a test user account





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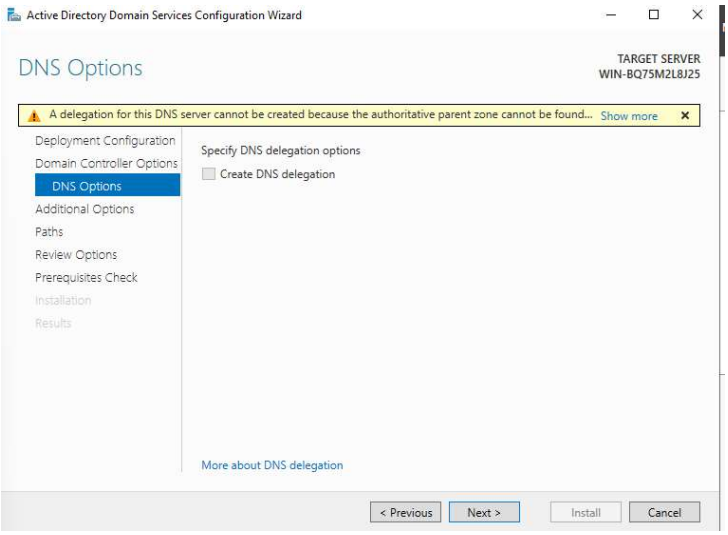
Problems/observations

- Got a warning about not having a proper static IP address when installing Active Directory Domain Servies, but I'm pretty sure I did that during day 1, so not going to worry about this warning for now, but keeping it in mind if it gives me issues in the future

- While I was configuring the DNS options for my new domain controller, I got an error message saying, “A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found”. I think this should be expected since I haven't setup a parent zone yet.



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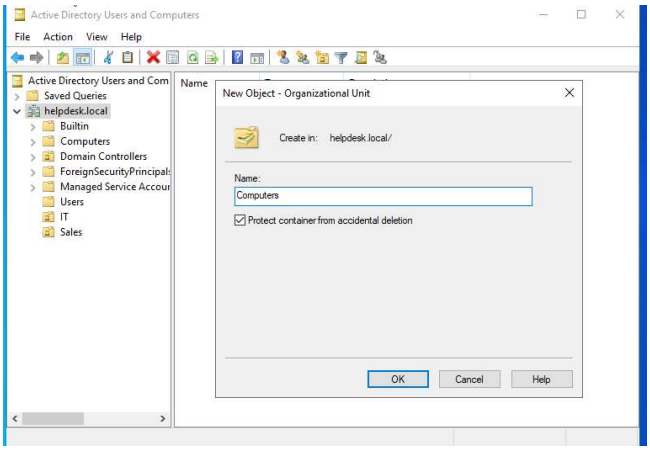
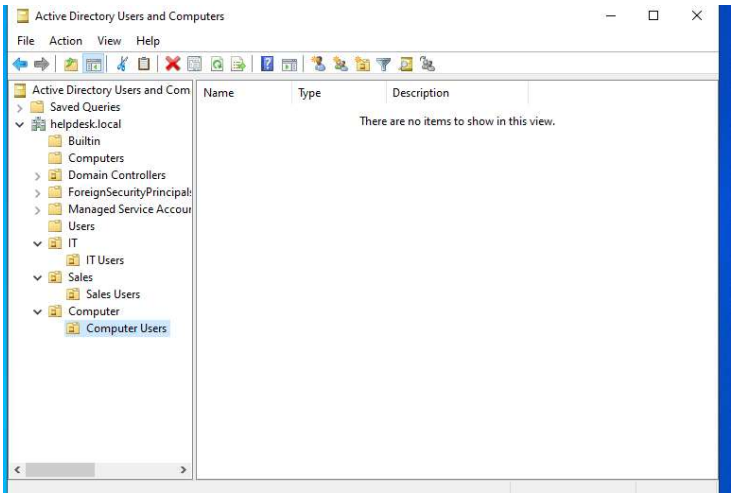
Day 3 Creating Organizational Units OU

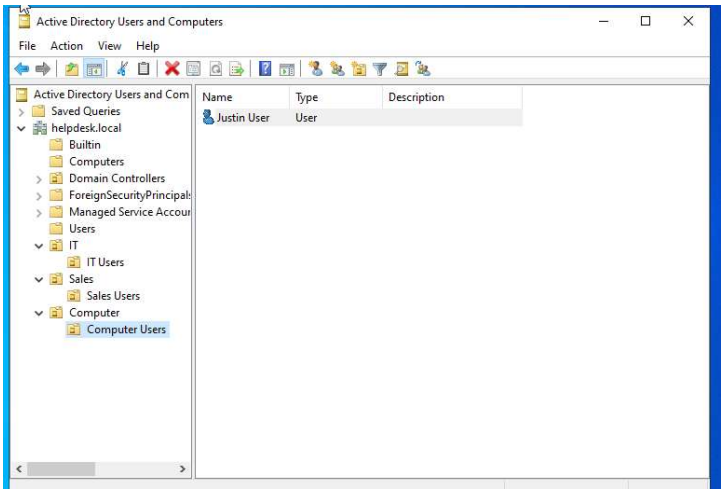
- Created 3 Organization Groups IT, Sales, and Computers



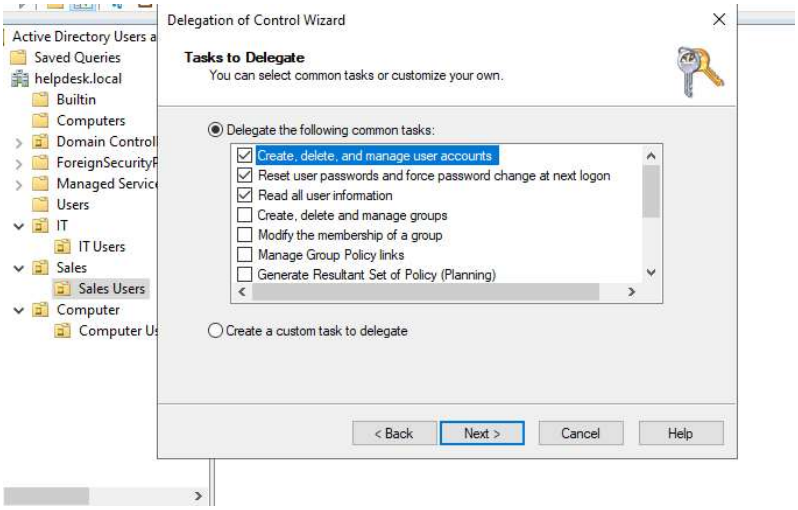
- Created a sub OU for each of the main OUs I just made



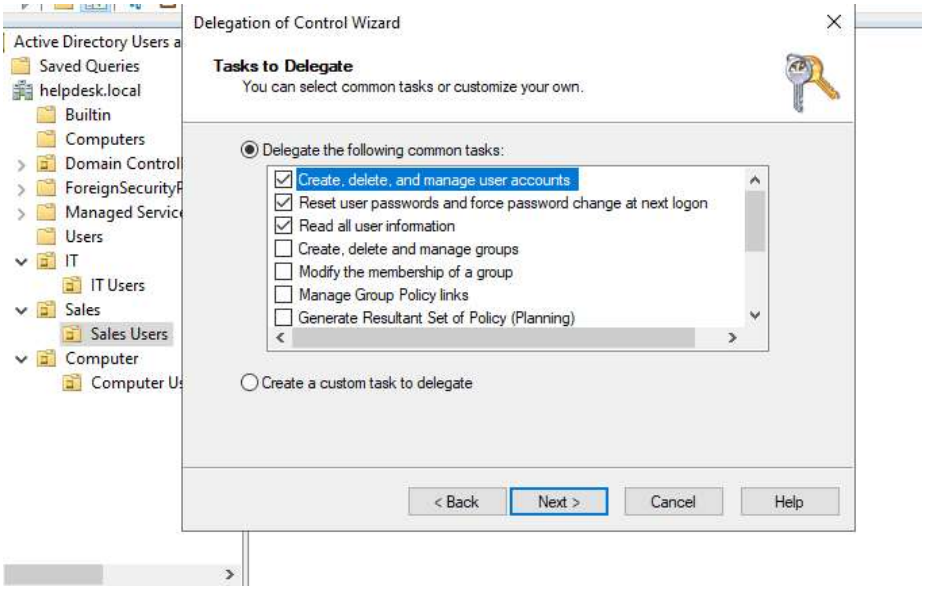
- Created a test user in each sub OU



- Delegating OU sales permissions to one of the Sales Users, so they have basic permissions to work with customers



- Screenshot of full OU structure under helpdesk.local showing both basic OU groups and thier subdirectories

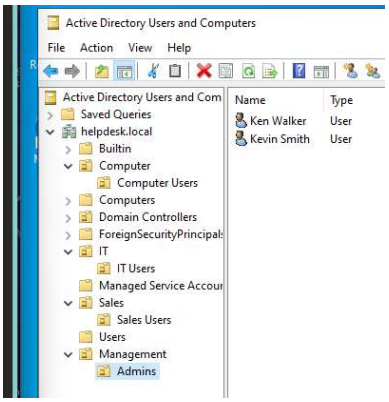


Problems/observations

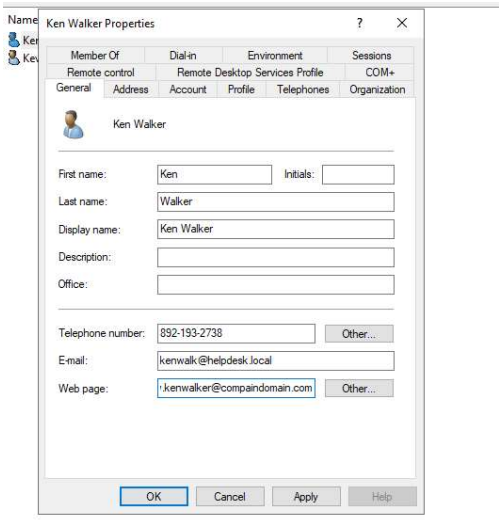
- Accidently made a group that already existed as Computers, so had to adjust that group name
- Other then that no issues, basic OU infrastructure is pretty straightforward

Day 4 Create a bunch of user accounts in the active AD

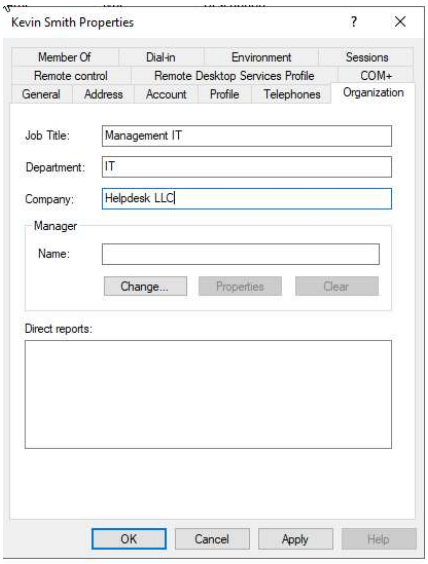
- Added 3-4 users to each of the IT, Sales, Computer departments



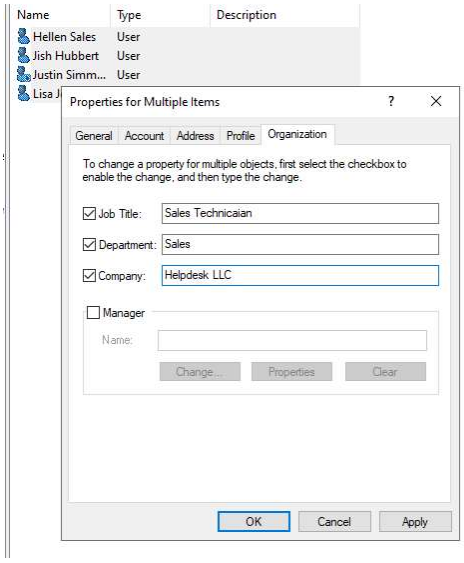
- Gave each newly created user a company phone number, company email address, and their own webpages



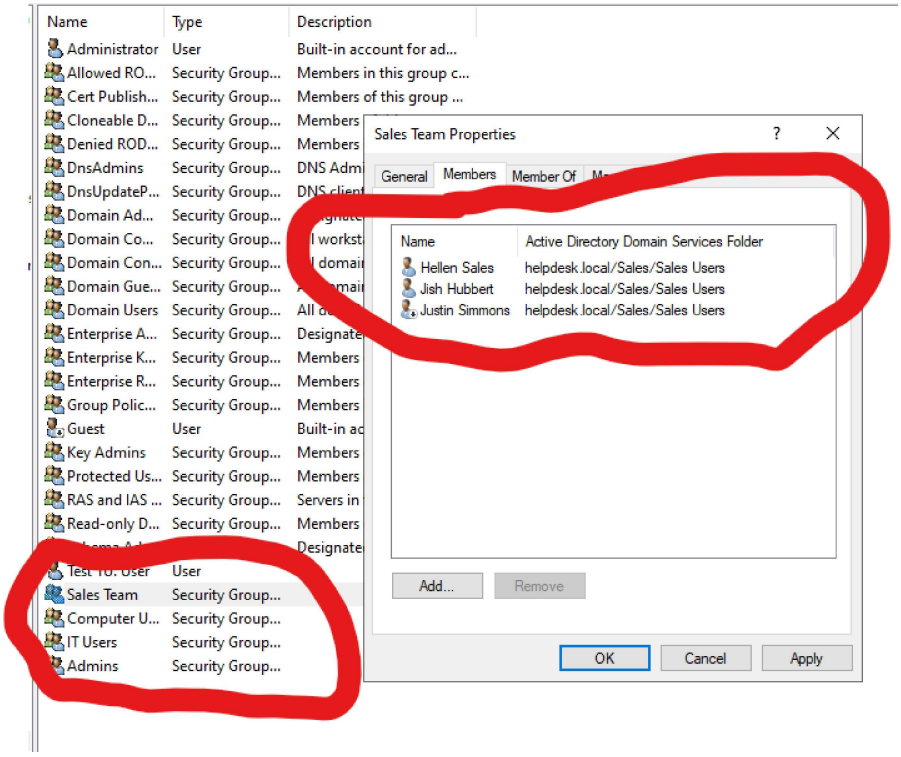
- Gave each user their own organization job titles, departments, and company name that will be used for Group Policy Management



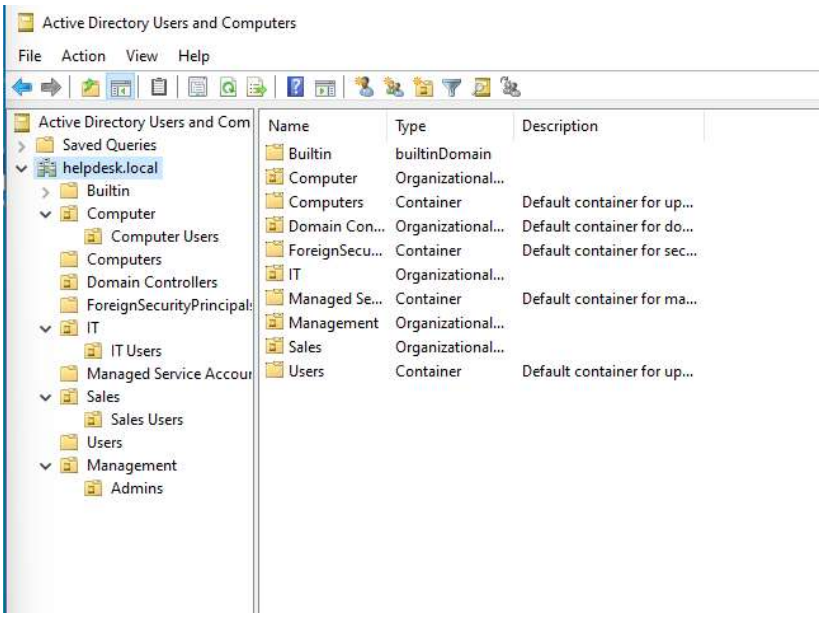
- While doing this, I used automation to assign multiple permissions at once



- Created a group object for each user group for security purposes, and then added each user in each group to it



- Updated OU structure

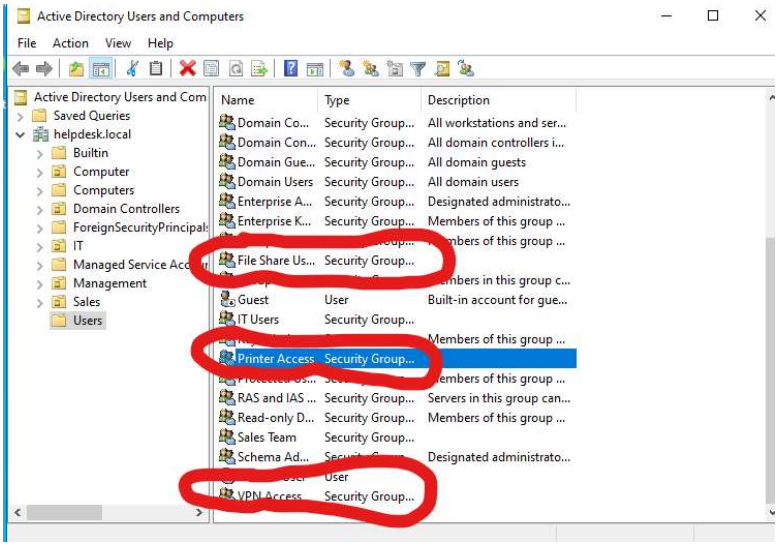


Problems/observations

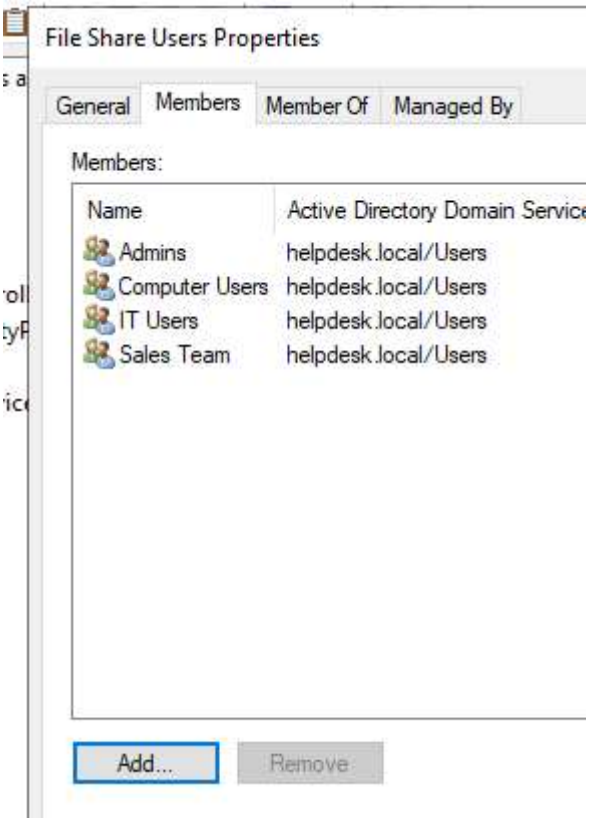
- Only real issue I had today was when I was adding users to the group objects, some of the users had the incorrect name, so the system couldn't find the object name

Day 5 Create Security Groups for each group each with its own seperate permissions

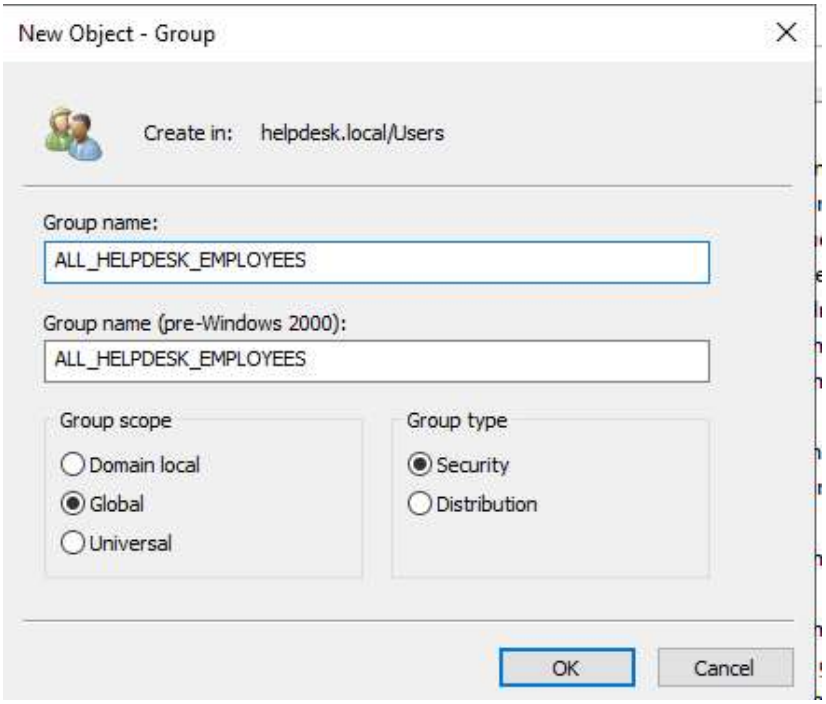
- Created 3 new security groups File Share Users, VPN Access, and Printer Access



- Gave all departments access to file share permissions, only gave Computer Users/IT Users printer access, and only gave VPN access to Admins



- Created a new nested group for all employees, so that we can put materials there that all employees need access too
- Also added Admin, Computer Users, IT Users, and Sales team



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Problems/observations

- No issues

Day 6 Group Policies

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Problems/observations

Day 7

Problems/observations